(i)

$$\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1}$$

$$\begin{aligned}\int_{1/m}^{m} \frac{x^2}{x+1}\,dx &= \left[\frac{x^2}{2} - x + \ln(x+1)\right]_{1/m}^{m} \\ &= \frac{m^2}{2} - m - \frac{1}{2m^2} + \frac{1}{m} + \ln\frac{m+1}{1+\frac{1}{m}} \\ &= \frac{(m-1)^3(m+1)}{2m^2} + \ln m\end{aligned}$$

(ii) Let $u = \dfrac{1}{x},\ dx = -\dfrac{1}{u^2}\,du$

$$\int_{1/m}^{m} \frac{1}{x^n(x+1)}\,dx = \int_{m}^{1/m} \frac{u^{n+1}}{u+1}\left(-\frac{1}{u^2}\right)\,du = \int_{1/m}^{m} \frac{u^{n-1}}{u+1}\,du$$

(iii)(a)

$$\begin{aligned}\int_{1/2}^{2} \frac{x^5+3}{x^3(x+1)}\,dx &= \int_{1/2}^{2} \frac{x^2}{x+1}\,dx + 3\int_{1/2}^{2} \frac{1}{x^3(x+1)}\,dx \\ &= \int_{1/2}^{2} \frac{x^2}{x+1}\,dx + 3\int_{1/2}^{2} \frac{x^2}{x+1}\,dx \\ &= 4\left(\frac{3}{8} + \ln 2\right) \\ &= \frac{3}{2} + 4\ln 2\end{aligned}$$

(iii)(b)

$$\int_{1}^{2} \frac{x^5+x^3+1}{x^3(x+1)}\,dx = \int_{1}^{2} \frac{x^2}{x+1}\,dx + \int_{1}^{2} \frac{1}{x+1}\,dx + \int_{1}^{2} \frac{1}{x^3(x+1)}\,dx$$

Let $u = 1/x,\ du = -x^{-2}dx = u^2dx$

$$\int_{1}^{2} \frac{1}{x^3(x+1)}\,dx = \int_{1/2}^{1} \frac{u^2}{u+1}\,du$$

$$\begin{aligned}\int_1^2 \frac{x^5+x^3+1}{x^3(x+1)}\,dx &= \int_{1/2}^2 \frac{x^2}{x+1}\,dx + \int_1^2 \frac{1}{x+1}\,dx \\ &= \frac{3}{8} + \ln 2 + \ln \frac{3}{2} \\ &= \frac{3}{8} + \ln 3\end{aligned}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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